

Presented by: Emna Ben Rejeb

GOLD PRICES FORECASTING

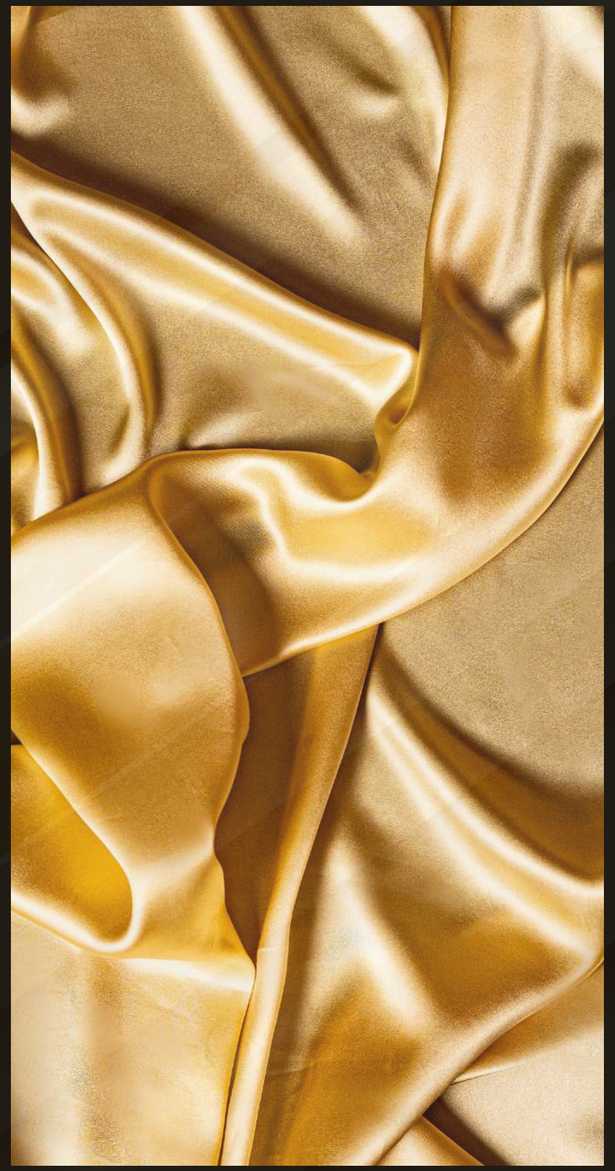
OUR DATASET

Description

Dataset: Daily gold prices
Time Range: January 1, 2014 – October 31, 2024
Raw Data Source: MCX Market

Variables

Date: Trading date
Price (Close Price): Final price at the end of the trading day
Open Price: Initial price at the start of the trading day
Highest Price: Maximum price reached during the day
Lowest Price: Minimum price recorded
Volume: Number of traded units
% Change: Percentage change from the previous closing price

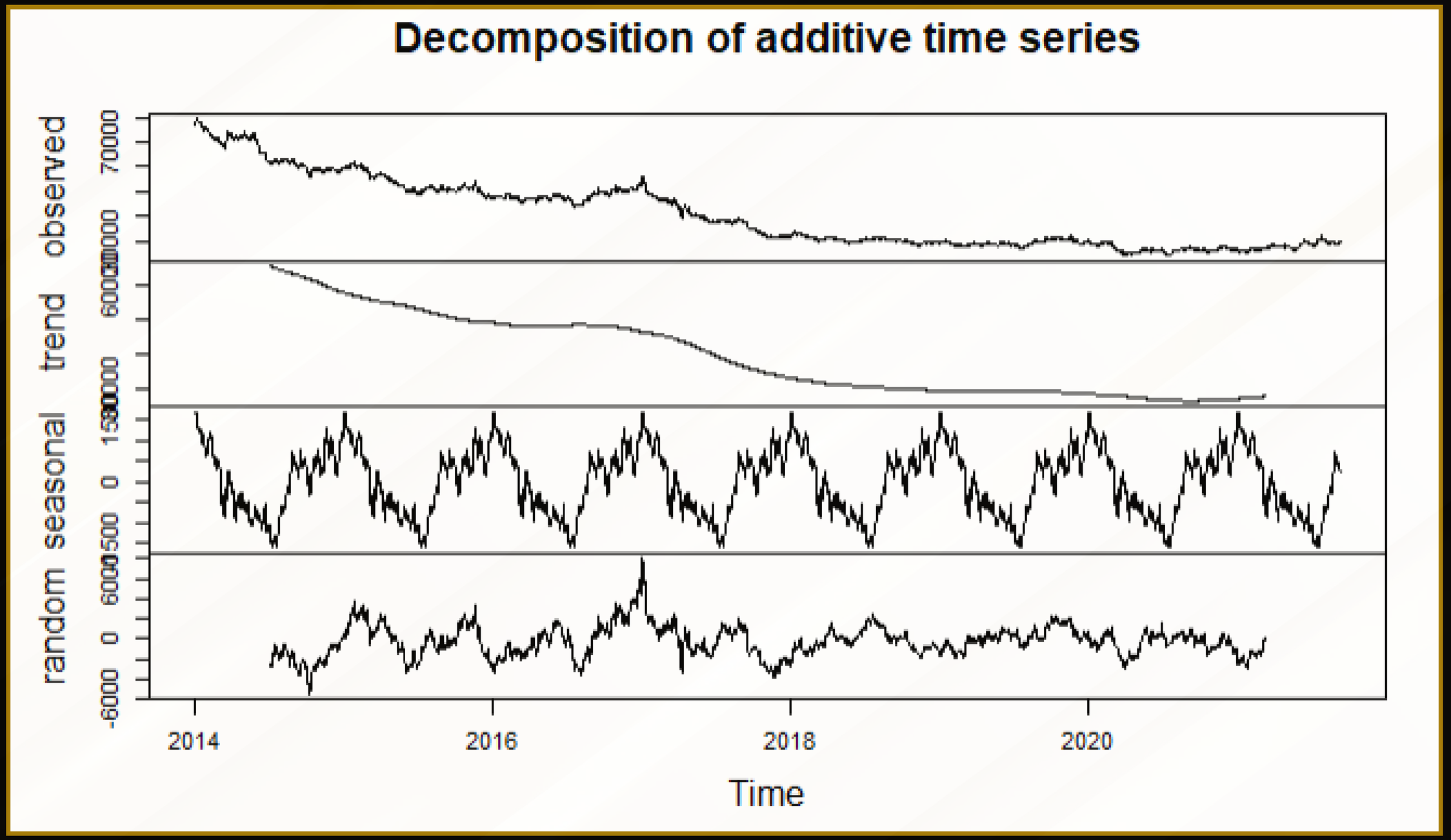


THE OBJECTIVE

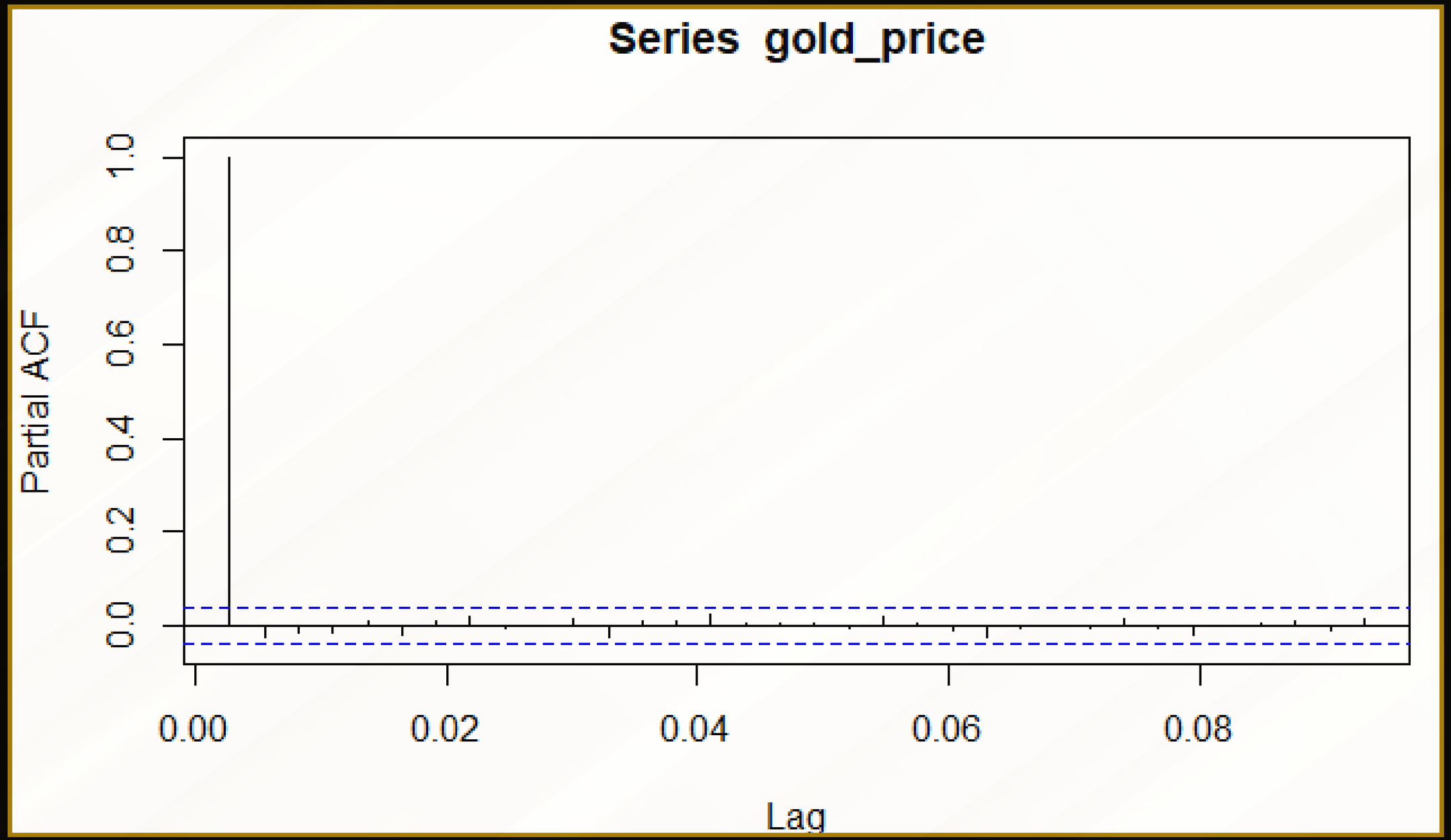
Variable of interest	Reasons
Close Price	Investment and risk management
	Economic and Market insights
	Business and Policy Decision-Making


Time Series Plots


Price Over Time
80000
60000
40000
Price
2014
2016
2018
2020
2022
Time


Decomposition of additive time series
observed
trend
seasonal
random
70000
60000
0
-6000
2014
2016
2018
2020
Time


Series gold_price
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.02
0.04
0.06
0.08
Lag


Series gold_price
Partial ACF
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.02
0.04
0.06
0.08
Lag

Stationarity



STATIONARITY TESTS

Dickey-Fuller test

Hypothesis:
H0:Time series is non-stationary
H1:Time series is stationary

P value: 0.5077

Conclusion: Gold prices data is non-stationary

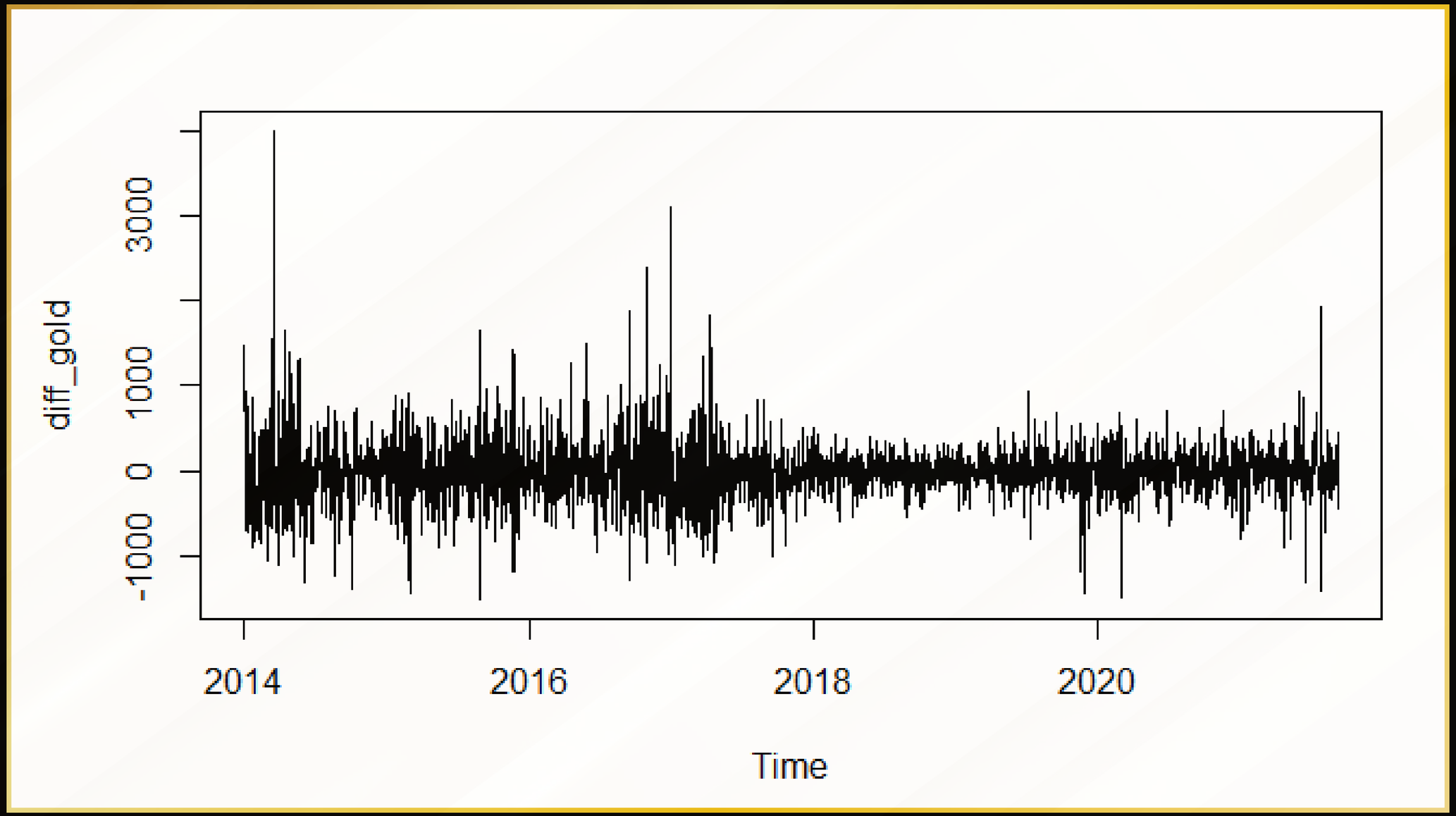
KPSS test

Hypothesis:
H0:Time series is stationary
H1:Time series is non-stationary

P value: 0.01

Conclusion: Gold prices data is non-stationary

DIFFERENCING TIME SERIES





Model Selection and Implementation

ARIMA MODEL: TRAINING SUBSET

Model Summary:

- ARIMA(4,0,5) with non-zero mean
- Log-likelihood = -16,437.78
- AIC = 32,897.55, BIC = 32,960.42

Coefficients & Significance:

- AR Terms: ar1 (-0.360), ar2 (1.3918), ar3 (0.2898), ar4 (-0.5791)
- MA Terms: ma1 (0.3429), ma2 (-1.3341), ma3 (-0.228), ma4 (0.5075), ma5 (-0.0572)
- Mean: -21.2640

Performance Metrics:

- Root Mean Squared Error (RMSE): 368.53
- Mean Absolute Error (MAE): 249.057
- Mean Absolute Scaled Error (MASE): 0.6866

SARIMA=ARIMA(4,0,5)

Market-Driven Fluctuations

Global Demand and Supply

Investor Sentiment and Speculation

ETS MODEL: (A,N,N)

Model Summary:

- **Smoothing Parameter (α): 0.0001**
- **Initial Level (l): -21.6187**
- **Sigma (Error Std Dev): 370.44**

Model Evaluation:

- AIC: 43843.33
- AICc: 43843.34
- BIC: 43860.48

Performance Metrics:

- Root Mean Squared Error (RMSE): 370.28
- Mean Absolute Error (MAE): 249.16
- Mean Absolute Scaled Error (MASE): 0.687



Model Evaluation

EVALUATION

ARIMA (4,0,5)	ETS (A,N,N)
RMSE= 368.5301	RMSE= 370.2751
AIC= 32897.55	AIC= 43843.33
BIC= 32960.42	BIC= 43860.48

THE BEST MODEL

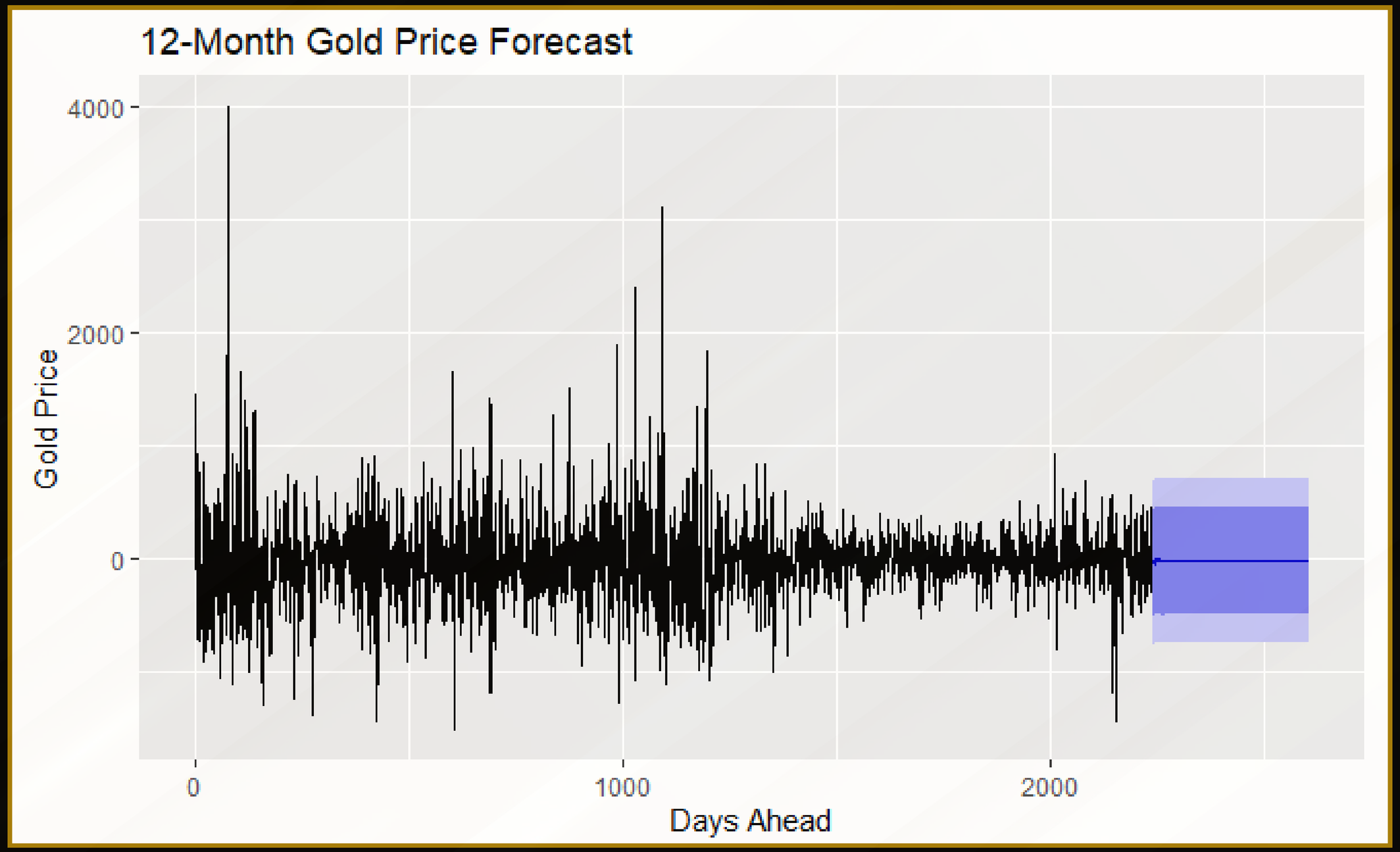
1

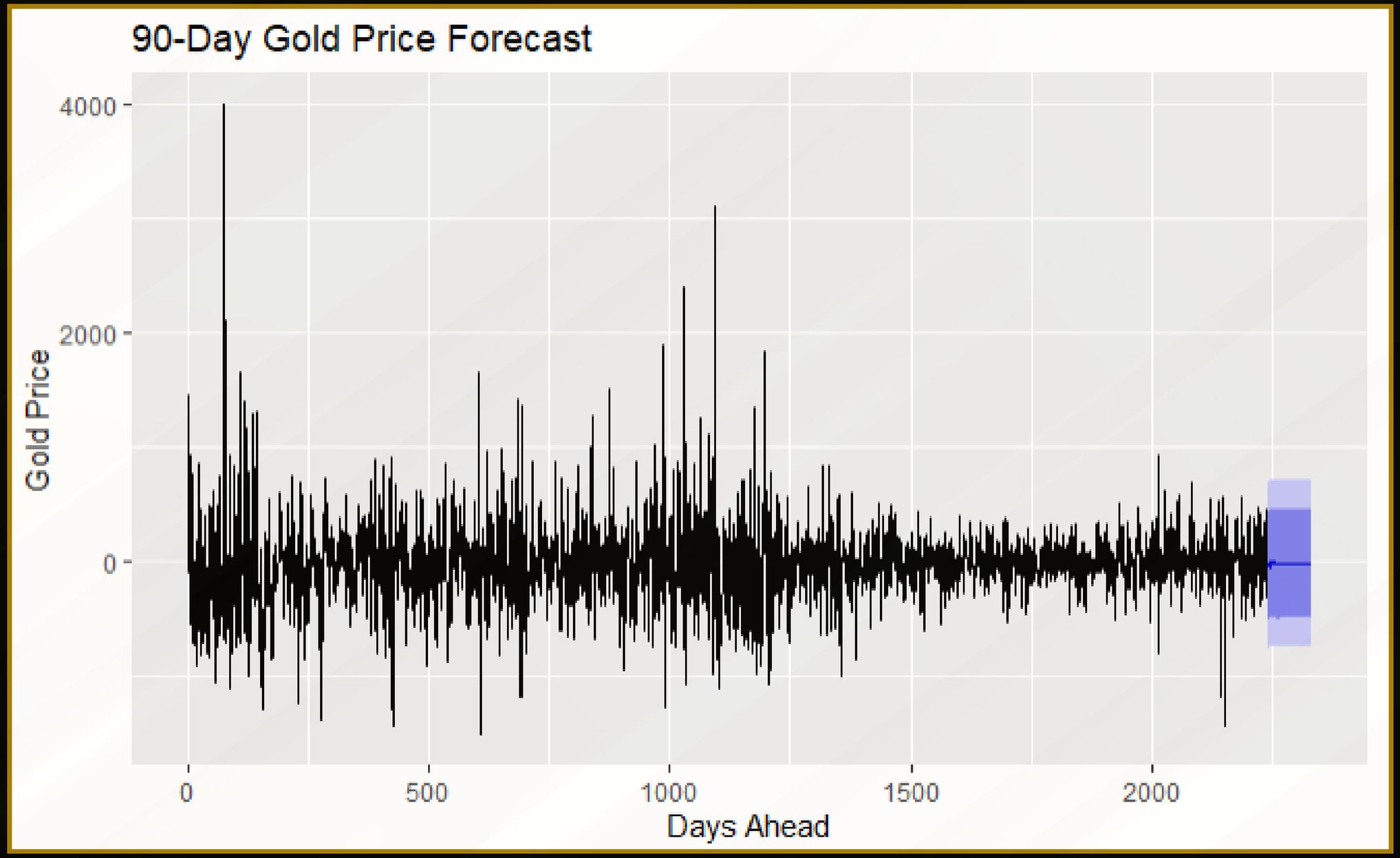
ARIMA (4,0,5)
Suitable for Forecasting

2

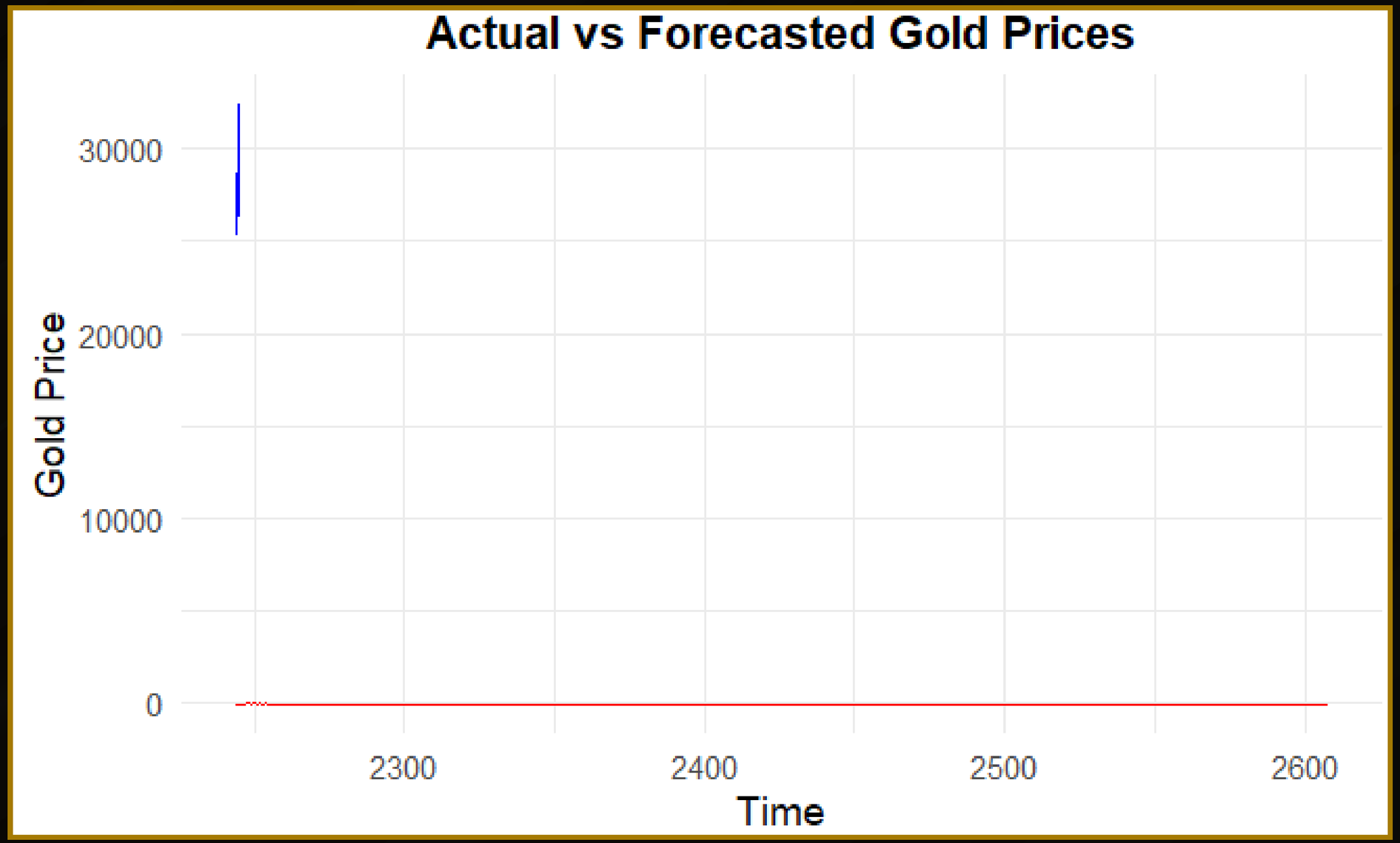
ETS (A,N,N)

Forecasting


12-Month Gold Price Forecast
4000
2000
0
Gold Price
0
1000
2000
Days Ahead


90-Day Gold Price Forecast
4000
2000
0
Gold Price
0
500
1000
1500
2000
Days Ahead

ACTUAL VS.FORECASTED




"There is still room for improvement, and further optimizations can enhance the accuracy of the forecast."

THANK YOU

